Shepherd Battery Model Parametrization for Battery Emulation in EV Charging Application

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***Abstract*—This paper deals with battery model parametrization in view of battery emulation applied to onboard electric vehicle (EV) chargers featuring efficient energy management system. For this purpose, accurate battery modeling and a good state of charge (SOC) estimation, are key issues. Hence Shepherd model is adopted as it corresponds to the one used by the target emulator and simulation environment PSIM. While the target emulator uses the original Shepherd model, the simulation environment uses the ameliorated model to enhance dynamics during battery current variation. A method to easily identify both models' parameters is described. Simulation results comparing the datasheet typical characteristic curve to the simulated curves from the adopted model show the compatibility of the model to the real battery cells' performance. Battery pack emulation presents good results while considering a Li-ion battery cell and as well as a battery pack for SOC higher than 20%.**

Keywords— Battery model, Lithium-ion battery, parameter identification, EV charging, State of Charge estimation

I. Introduction

During the last decades, many countries in the world targeted lower gas emissions in the transportation sector which involved a fast growth of electric vehicles and storage systems markets. As Battery Energy Storage System (BESS) is a key element in electric vehicle development, many researchers have focused their attention on the improvement of the energy management system for a more reliable and efficient operation. Mainly three battery types can be found in EVs: Lead-Acid battery, Nickel-metal hydride battery (NiMH), and lithium-ion battery. However, the most attractive solution for pure Electric Vehicles (EV), Hybrid Electric Vehicles (HEV), and Plug-in Hybrid Electric Vehicles (PHEV) is Li-ion batteries which present the major advantages of higher energy density and higher output voltage than the two other types. These batteries are recyclable, have a longer lifetime and their cost is continuously decreasing [1], [2].

To ensure a safer and more efficient battery system with a longer lifetime, Battery Management System (BMS) is an essential part. Therefore, choosing the appropriate battery model, identifying its parameters, and estimating the battery State of Charge (SOC) are the three initial steps required to design and implement the BMS and prevent hazardous battery operation [3]-[5]. Many papers focused on battery modeling technologies and proposed a classification of these methods such as in [1], [6], [7], and [8]. Authors in [1] divided battery models into four categories, detailing each method's features and giving the advantages and disadvantages of each one: empirical model, data-driven model, electrochemical model, and Equivalent Circuit Model (ECM). However, authors in [6] have only considered the three last categories as they are the most popular ones whereas only the last two methods are discussed and reviewed by [7]. The next step after choosing the battery model is the identification of its parameters. Authors in [7] proceeded to the identification of seven typical battery models using recursive least squares method with an optimal forgetting factor. The seven methods considered in this paper are the following: Shepherd model, Unnewehr Universal model, Nernst model, and Combined model as simplified electrochemical models, Rint model, Thevenin model, and DP model as equivalent battery circuit models.

The state of charge (SOC) of a battery is the amount of available energy which is difficult to be measured. Hence, SOC estimation of Li-ion cells for EV application is an important research issue because the SOC parameter is required for battery life cycle and power prediction. By accurately estimating the Li-ion battery SOC, online SOC estimation can be performed. Thereby, both BMS efficiency and battery safety can be highly improved for EV charging systems [4].

Among these methods, the most commonly used are the Open-Circuit-Voltage method, ampere-hour integration (Ah) method, neural network, and Extended Kalman filter (EKF) [4] and [9]. These methods are reviewed and analyzed according to their advantages and disadvantages in many research papers [2], [4], and [6]. In [4], authors proposed a new technique including both Extended Kalman Filtering (EKF) and ampere-hour (Ah) integration method in order to improve the SOC accuracy under different experimental conditions.

This paper deals with a systematic approach of lithium-ion battery pack parameters identification for electric vehicles. Shepherd model and modified Shepherd model are used to implement the battery parameters into a *15kVA* battery emulator B2C from Cinergia firm. In fact, this B2C battery emulator integrates Shepherd mathematical model to emulate real batteries from different technologies such as Li-ion, NiMH, NiCd, Pb, etc. Hence, it is necessary to identify each battery parameter so that to emulate it using the B2C. Given the discharging characteristic of Li-ion cells from the datasheet, the proposed method can be applied to any type of battery technology. In this paper, only Li-ion technology is discussed as this is the most common one for EVs. In order to validate the identifying parameters method, the model is compared to the datasheet discharging cell characteristic. Another comparison was conducted using PSIM software battery model and implemented on the B2C battery emulator. The main contributions of this paper are listed below:

- Li-ion battery cell model is established, and its parameters are identified for two different Li-ion battery cells: VL 34570 and Pylontech cell.

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- The model is compared to PSIM software battery model and to the datasheet data based on the modified Shepherd model.
- Battery parameters are identified using Shepherd model and implemented on the B2C battery emulator for both VL 34570 and Pylontech cells and also for two packs based on these cell types.
- Battery cells and battery packs characteristics using either PSIM software model, modified shepherd model, or original shepherd model are compared and discussed.

This paper is structured as follows. In section II, onboard EV charging system based on a half-bridge DC/DC converter with the charging/discharging control are described. Section III presents Shepherd model and modified Shepherd model for two Li-ion battery cells as well as their parameters identification method. To validate these models, simulation results using PSIM software are performed and then compared to the datasheet characteristics of each Li-ion cell in section IV. In addition to that, the battery model implemented on a B2C battery emulator is also compared to the previous models for two battery cells and two battery packs. Finally, the main findings and conclusions of this work are summed up in section V.

II. Onboard EV Charger System

The targeted application of this paper is the Battery Management System of an onboard EV charger. In fact, EV chargers can be classified as off-board chargers and on-board chargers and the power flow can be unidirectional or bidirectional. On-board chargers are integrated into the EV and so their charging power is limited due to space and weight constraints. However, off-board chargers are installed in charging stations and have high power levels (from *50 kW* to *175 kW*) and can be three-phase AC or DC charging types and it is considered as fast charging [10], [11].

Typical simplified block diagram for onboard charger is illustrated in Fig. 1. This charger includes an AC/DC boost converter playing the role of Power Factor Corrector (PFC) and a non-isolated DC/DC converter followed by an output LC filter to charge the EV battery.

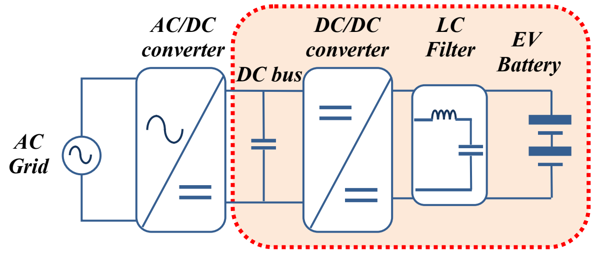

Fig. 1. Onboard EV charger block diagram

In this paper, we will consider that the DC bus voltage is well regulated through the AC/DC converter [12], at the level of 500V and we will focus on the left part of the onboard charger (in red color). The DC/DC converter power and control parts are described in Fig. 2.

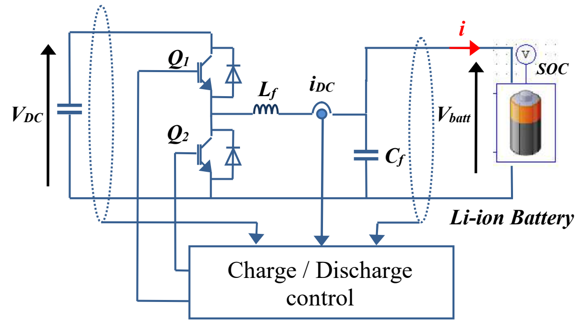

Fig. 2. DC/DC converter topology

It consists of IGBT based bidirectional buck/boost converter with an LC output filter. The Li-ion EV battery pack model is used from PSIM software library and its parameters are given in TABLE I. Battery cells are VL 34570 type. The charging control can be conducted in Constant Current/ Constant Voltage (CC/CV) mode while discharging can be done under constant current mode; both controls require an accurate SOC estimation. Fig. 3 and Fig. 4 illustrate both charging and discharging operations for the VL 34570 battery pack with a nominal capacity of *64.8 Ah*. To simplify the system in Fig. 2, the converter was replaced by an independent current source. Fig. 3 shows the charging process from an initial SOC equal to zero until reaching *1* under a constant current of *13.2 A*. Then, constant voltage mode is applied. It can be seen at Fig. 3 that the battery pack capacity is *64.8 Ah* when SOC equals *1*. Discharging the battery pack is depicted in Fig. 4 under a constant current of *-13.2 A,* and showing the SOC evolution from *1* to *0*.

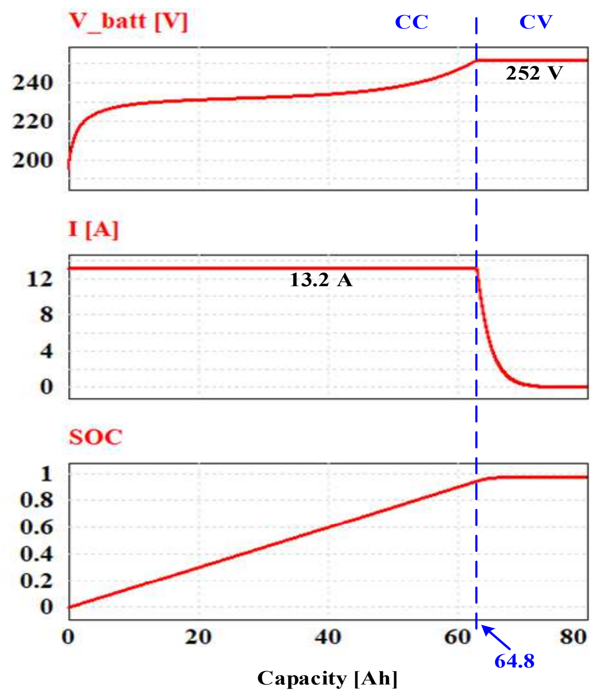

Fig. 3. Battery charging curves

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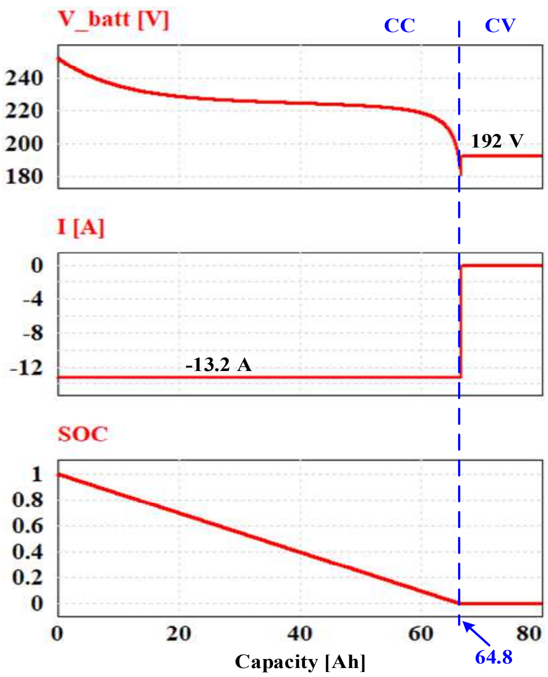

Fig. 4. Battery discharging curves

III. EV BATTERY MODELING

A. Shepherd battery model

The Li-ion EV battery circuit model from PSIM library uses the modified Shepherd mathematical model. The Shepherd battery model discharge equation [13], from which originated the modified model is given by (1):

$$V_{batt} = V_0 - K\left(\frac{Q}{Q-it}\right)i - R_0\, i + Ae^{-(B\,it)} \qquad (1)$$

Where V_{batt} is the actual battery voltage, V_0 its open circuit voltage at full capacity, K its polarization resistance coefficient, Q its capacity, i its actual current, R_0 its internal resistance, A its exponential zone amplitude, B its exponential zone time constant and it the removed charge. Therefore, the B2C battery emulator parametrization goes through the determination of these parameters.

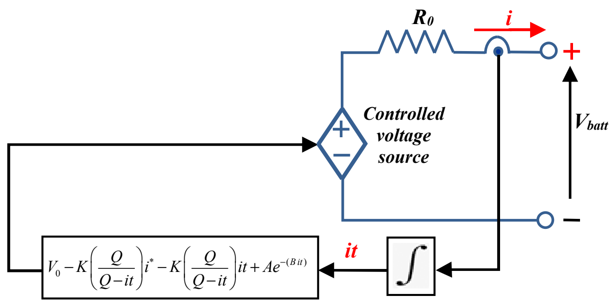

Fig. 5. Modified Shepherd battery model

Although the battery model expressed in (1) has been largely used, it has the disadvantage of being unable to accurately represent the voltage dynamics when the current varies. Therefore, an ameliorated version of Shepherd model, taking into account the open circuit voltage as a function of the state of charge (SOC) as well as being able to represent the voltage dynamics during current variation has been proposed in [14].

The ameliorated Shepherd model is presented in Fig. 5. Discharge equation of this model is then given by (2), where i^* represents the filtered current.

$$V_{batt} = V_0 - K\left(\frac{Q}{Q-it}\right)i^* - K\left(\frac{Q}{Q-it}\right)it - R_0\, i + Ae^{-(B\,it)} \qquad (2)$$

In this equation, the term in i^* represents the polarization resistance while the one in it is the polarization voltage.

For both models, the state of charge estimation is expressed in (3), where SOC_{init} is the initial SOC.

$$SOC = SOC_{init} - \frac{1}{Q}\int i\,dt \qquad (3)$$

B. Battery parametrization

It consists in determining the constitutive parameters of the discharge equations in (1) and (2), namely, V_0, K, Q, R_0, A, and B. Some of these parameters such as Q and R_o are readily available from the basic parameters table from the manufacturer datasheet while others are calculated using data from the typical discharge curve provided in the datasheet. For this purpose, four points are drawn from the discharge curve, namely the point of fully charged voltage, the point of the end of the exponential zone, the point of the end of the nominal zone, and the point of the maximum capacity as illustrated in Fig. 6. The discharge curve characteristic is given for a constant current discharge I.

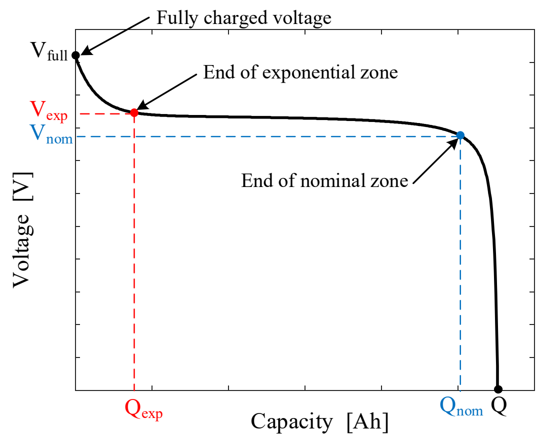

Fig. 6. Typical battery discharge curve

The parameters are determined by solving a system of equations using the four points from the datasheet's characteristic discharge curve. Considering (1), at the beginning of the curve characteristic, the voltage corresponds to the fully charged voltage, and the supplied current it is null. Hence, the battery voltage at fully charged state is expressed in (4):

$$V_{full} = V_0 - K\,I - R_0\,I + A \qquad (4)$$

At the point of the end of the exponential zone, the battery voltage is given by (5):

$$V_{exp} = V_0 - K\left(\frac{Q}{Q-Q_{exp}}\right)I - R_0\,I + Ae^{-(B\,Q_{exp})} \qquad (5)$$

Likewise, the battery voltage at the point of the end of the nominal zone is given in (6).

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$$V_{nom} = V_0 - K\left(\frac{Q}{Q - Q_{nom}}\right) I - R_0\, I + A e^{-(B\, Q_{nom})} \quad (6)$$

The parameter B can be approximated to a given value according to the shape of the discharge current, in other words, the battery cell technology [8] and given by (7):

$$B = \frac{n}{Q_{\exp}} \quad (7)$$

For a lithium iron phosphate (LFP) battery cell, n is approximated to *4* while for a nickel manganese cobalt (NMC) one it is approximated to *2*.

For the original Shepherd battery model, the parameters V_0, K, A, and B are determined by resolving the system of equations in (8).

$$\begin{cases} V_{full} = V_0 - K\, I - R_0\, I + A \\ V_{\exp} = V_0 - K\left(\frac{Q}{Q - Q_{\exp}}\right) I - R_0\, I + A e^{-(B\, Q_{\exp})} \\ V_{nom} = V_0 - K\left(\frac{Q}{Q - Q_{nom}}\right) I - R_0\, I + A e^{-(B\, Q_{nom})} \\ B = \frac{n}{Q_{\exp}} \end{cases} \quad (8)$$

Considering the modified Shepherd battery model in (2), the system of equations to be solved for the parameter determination becomes the one expressed in (9).

$$\begin{cases} V_{full} = V_0 - R_0\, I + A \\ V_{\exp} = V_0 - K\left(\frac{Q}{Q - Q_{\exp}}\right)\left(I + Q_{\exp}\right) - R_0\, I + A e^{-(B\, Q_{\exp})} \\ V_{nom} = V_0 - K\left(\frac{Q}{Q - Q_{nom}}\right)\left(I + Q_{nom}\right) - R_0\, I + A e^{-(B\, Q_{nom})} \\ B = \frac{n}{Q_{\exp}} \end{cases} \quad (9)$$

IV. Simulation Results & Discussion

A. *Model validation*

Simulations considering two battery cells are conducted in this section.

The first cell is the one used for the onboard charger used in section II, referenced as VL 34570. It will be referred to as *cell1* in the following. The second cell, referred to as *cell2,* corresponds to the one used as battery storage system available in the microgrid platform of the lab. It is referenced as "Energy Cell 25N" from Pylontech. The parameters of the battery pack for a typical discharge curve at *C/5* rate for *cell1* and *C/2* for *cell2* are presented in TABLE I.

Since the model used by the PSIM software corresponds to the Shepherd modified model, this latter is used first to generate a simulated curve whose characteristic is the closest to the one of the datasheets. The simulated curve is fitted by adjusting the end of exponential and nominal zone points to have the same as the one provided by the battery datasheet.

TABLE I. BATTERY PACKS PARAMETERS

Parameters		VL 34570	Energy cell 25N
Battery pack information	Number of cells in series	60	15
	Number of cells in parallel	12	2
From datasheet	Rated voltage [V]	3.7	3.2
	Discharge cut-off voltage [V]	2.5	2.5
	Rated capacity [Ah]	5.4	25
	Internal resistance [Ω]	0.05	0.0012
From datasheet discharge curve	Full battery voltage [V]	4.2	3.58
	Exponential point voltage [V]	3.75	3.27
	Nominal point voltage [V]	3.6	3.13
	Maximum capacity [Ah]	5.56	26.1
	Exponential point capacity [Ah]	2.5	0.44
	Nominal point capacity [Ah]	5.2	21.31

Both original and modified Shepherd models' simulation results are compared to the datasheet characteristic curve. The three curves for modified model, Shepherd model and datasheet one are presented in Fig. 7 and Fig. 8 for *cell1* and *cell2* respectively. For *cell1*, discharging profile for a discharge current of *1.08 A* (*C/5 rate*) at *+20°C* is considered. As for cell2, discharging profile for a discharge current of *12.5 A* (C/2) at *+20°C* is considered.

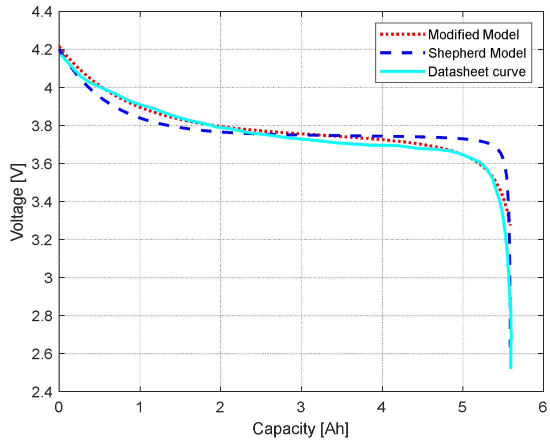

Fig. 7. Comparison between original and modified Shepherd model with datasheet curve for VL 34570 cell

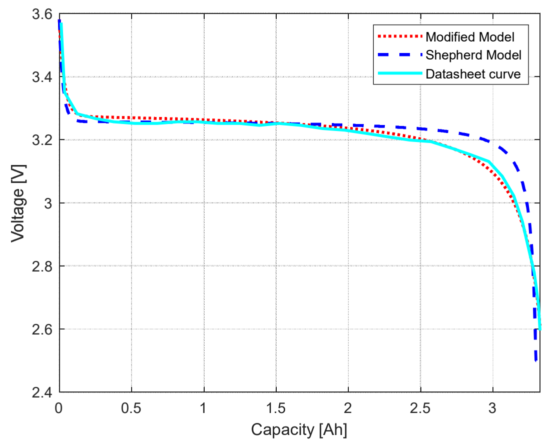

Fig. 8. Comparison between original and modified Shepherd model with datasheet curve for Pylontech cell

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As expected, for both cells, the modified Shepherd model curve is almost identical to the one of the datasheets while the original model one is slightly shifted near the point of exponential and nominal zones. This is due to the poor dynamic representation of the original Shepherd model.

B. Battery emulation

Once the parameters are determined, the battery cells and their packs are emulated using a bidirectional converter *B2C* from Cinergia. This emulator is a *15 kVA* DC programmable power supply that can operate as a battery charger, battery emulator, or solar panel emulator, in addition to generate controlled DC source or load. In this work, the B2C is used as a battery emulator. B2C emulator features a dedicated software application running on a host computer, enabling its configuration according to the user's need. The battery emulation environment is illustrated in Fig. 9.

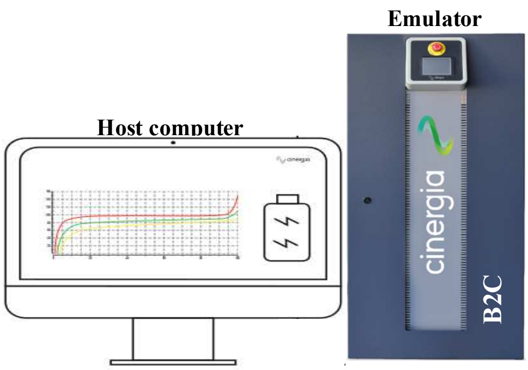

Fig. 9. Battery emulation environment

This emulator uses the original Shepherd model; thus, the corresponding parameters are introduced through the application interface as illustrated in Fig. 10. These parameters allow setting up the emulator operating point once loaded. They are the open circuit voltage, the polarization gain, the maximum capacity, the exponential coefficients, and the internal resistance, in box A of Fig. 10.

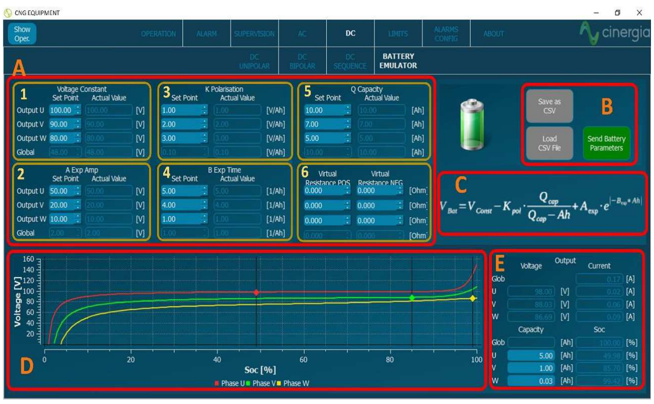

Fig. 10. B2C Emulator interface [15]

Comparison while considering the batteries from TABLE I. is provided in the following. With the series-parallel association of the cells, battery *pack1* has a rated capacity of *64.8 Ah* and a rated voltage of *222V* while battery *pack2* has a rated capacity of *50 Ah* and a rated voltage of *48V*.

As the B2C emulator only provides battery voltage versus SOC curve as can be seen in box D of Fig. 10, the comparison to the simulated curves is performed on the voltage versus SOC. Fig. 11 and Fig. 12 illustrate the comparison between the original and modified Shepherd model curve to the one provided by the offline simulation of the B2C emulator considering cell1 and cell2. On the other hand, Fig. 13 and Fig. 14 show the same comparison as in Fig. 11 and Fig. 12 while considering battery *pack1* and battery *pack2* respectively.

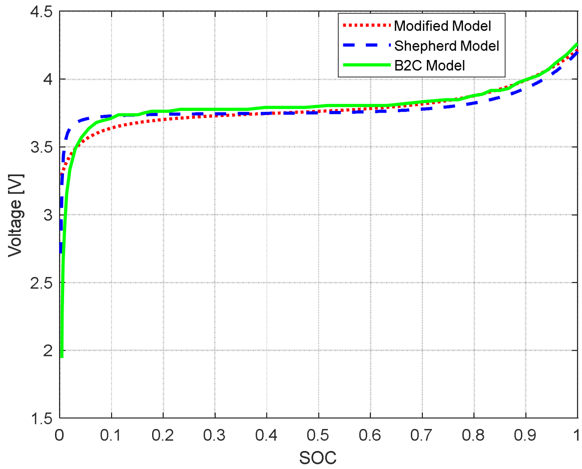

Fig. 11. Comparison between original and modified Shepherd model with B2C model for VL 34570 cell

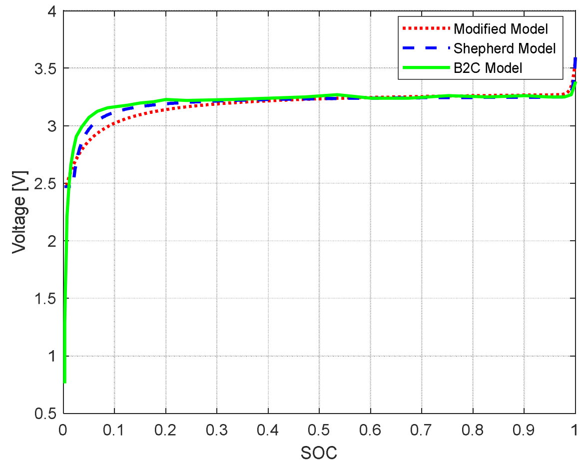

Fig. 12. Comparison between original and modified Shepherd model with B2C model for Pylontech cell

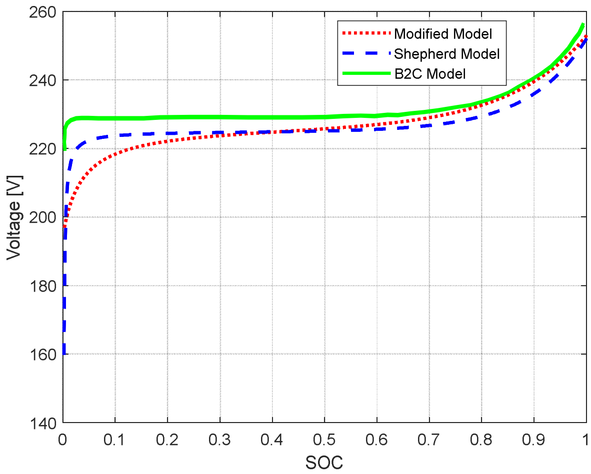

Fig. 13. Comparison between original and modified Shepherd model with B2C model for VL 34570 battery pack

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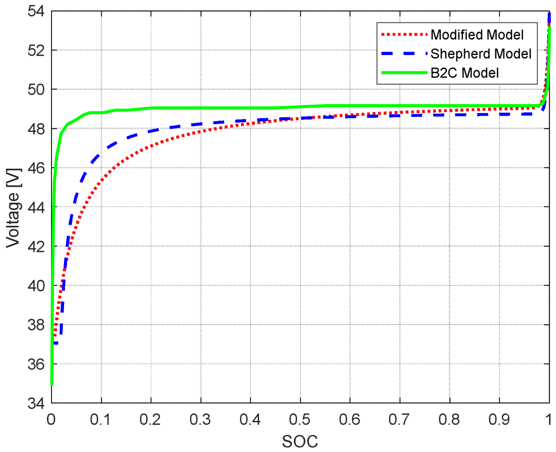

Fig. 14. Comparison between original and modified Shepherd model with B2C model for Pylontech pack

Considering the battery cells from Fig. 11 and Fig. 12, it can be concluded that the original Shepherd model and the B2C curves are almost identical, except at the beginning of the curves. This is explained by the fact that the emulator parameters are rounded to the nearest tenth, except for the internal resistance to the nearest hundredth. Since the polarization coefficient is of the order of 10^{-3}, it is rounded to 10^{-2} while entering the emulator parameter.

Likewise, for the battery packs in Fig. 13 and Fig. 14, the B2C simulation result and the original Shepherd model are similar in shape with a slight mismatch, especially at the beginning of the curve. The error due to rounding to the nearest tenth is even more important with the association of cells in pack. For a state of charge higher than *20%,* the B2C emulator result is fairly acceptable.

V. Conclusion

This paper deals with the Li-ion EV battery model for onboard charger control implementation. The application type and its objectives have been presented first. Then the modelling required for the battery emulation has been detailed considering Shepherd battery model. The original Shepherd model as well as the modified model has been elaborated with their corresponding parameters' determination methodology. Simulation comparing the discharge curve from datasheet and simulated curves from both Shepherd models demonstrated the efficiency of the proposed identification battery model parameters. The modified Shepherd model is closer to the actual characteristic curve of the battery cells than the original one, which makes the first a good model for simulation study. However, the original model is the one required by the B2C emulator. Hence, the battery emulation results show a slight difference from simulation results because of the, but they remain acceptable for SOC higher than *20 %.*

As perspectives, the battery model and its SOC estimation having been addressed, the implementation of the onboard EV charger system will be realized using real-time simulation control run on OPAL-RT, while using the B2C battery emulator as a battery.

Acknowledgment

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